**Definition**

If a line touches the given circle at only one point, then it is called tangent to the circle.

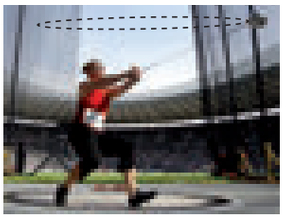
## Real life examples of tangents to circles

(i) When a cycle moves along a road, then the road becomes the tangent at each point when the wheels rolls on it.



Fig. 4.53(a)

(ii) When a stone is tied at one end of a string and is rotated from the other end, then the stone will describe a circle. If we suddenly stop the motion, the stone will go in a direction tangential to the circular motion.



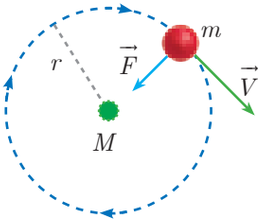



Fig. 4.53(b)

## Some results on circles and tangents

1. A tangent at any point on a circle and the radius through the point are perpendicular to each other.

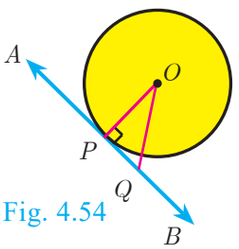
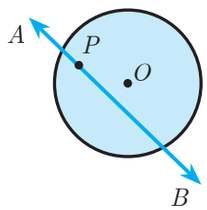
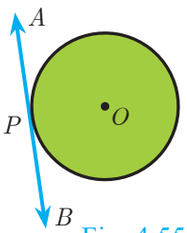
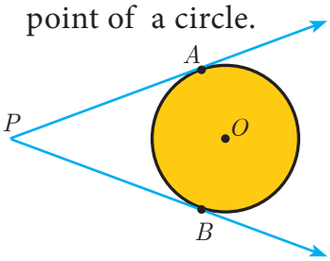



Fig. 4.54

| 2. (a) No tangent can be drawn from an interior point of the circle. | (b) Only one tangent can be drawn at any point on a circle. | (c) Two tangents can be drawn from any exterior point of a circle. |
|---|---|---|
|   Fig. 4.55(a) |   Fig. 4.55(b) |   Fig. 4.55(c) |

3. The lengths of the two tangents drawn from an exterior point to a circle are equal,

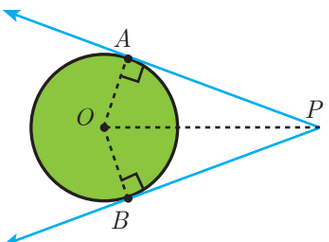



Fig. 4.56

**Proof :** By 1. $OA \perp PA, OB \perp PB$. Also $OA = OB =$ radius, $OP$ is common side. $\angle AOP = \angle BOP$

Therefore, by SAS Rule $\Delta OAP \cong \Delta OBP$. Hence $PA = PB$

4. If two circles touch externally the distance between their centers is equal to the sum of their radii, that is $OP = r_1 + r_2$

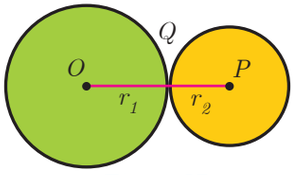



Fig. 4.57

**Proof :** Let two circles with centers at $O$ and $P$ touch other at $Q$.

Let $OQ = r_1$ and $PQ = r_2$ and let $r_1 > r_2$.



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